

Updated Sep 2026

EDUCATION

Purdue University

PhD in Mathematics

West Lafayette, IN

since August 2025

California Polytechnic State University

M.S. in Mathematics (Applied Math Specialization, with Distinction)

San Luis Obispo, CA

June 2025

- Advisor: Dana Paquin
- Thesis Title: Stability Insights from Modeling Chronic Myelogenous Leukemia

California Polytechnic State University

B.S. in Mathematics (Applied Math Concentration)

San Luis Obispo, CA

June 2023

- Minors in Statistics, Anthropology & Geography

RESEARCH EXPERIENCE

Graduate Research Assistant — Supervisor: Dana Paquin

Project Title: Mathematically Modeling Chronic Myelogenous Leukemia

June 2024 - June 2025

- Identified computable time to remission in MATLAB of delay-differential equation system describing CML dynamics
- Derived equilibrium solutions for the system, representing cancer-free and cancer-immune coexistence environments, and determined parameter restrictions for the asymptotic stability of each

Undergraduate Research Assistant — Supervisor: Ryan Tully-Doyle

Project Title: Graph-Theoretic Interpretations of the Nevanlinna Representation

June 2023 - Sep 2023

- Coded Mathematica functions to generate graph products, adjacency matrices and representing functions that track possible paths
- Proved two-variable representing function results for graphs joined at common vertex

Undergraduate Senior Project — Advisor: Elena Dimitrova

Project Title: Exploring Community Structure in Online Discourse with Python

Apr 2023 - Sep 2023

- Ran Louvain algorithm on mention-based network from vaccination-related tweets with NetworkX, and calculated degree centrality across communities
- Used sentiment analyzer from nltk library in Python to quantify and compare tweet sentiment within communities

BEACoN Research Scholar — Supervisor: Joyce Lin

Project Title: Using Machine Learning to Improve Targeting for Farmer Training

Jan 2023 - June 2023

- Used Pandas and NumPy libraries in Python to clean Cambodian survey data (baseline and endline), construct income features

Undergraduate Research Assistant — Supervisor: Jeffrey Liese

Project Title: Characterizing Non-Singular Graphs with Games

June 2022 - Sep 2022

- Calculated optimal strategy vectors using Mathematica, by viewing adjacency matrices as two-player, zero-sum matrix games; proved non-singularity for various graph families

PAPERS

Tully-Doyle, R., Adlin, L., **Thai, G.**, Tiscareno, S. (2026) *Cauchy Transforms of Colored Graphs in Two Variables*. *Linear Algebra and its Applications*, 2026 (731): pp. 1-21. <https://doi.org/10.1016/j.laa.2025.11.002>.

Paquin, D., Gross, L., Stewart, A., **Thai, G.** (2025) *Numerical Analysis of Critical Values for Remission During Imatinib Treatment of Chronic Myelogenous Leukemia*. *Mathematical Biosciences and Engineering*, 2025 22(6): pp. 1551-1571. <https://doi.org/10.3934/mbe.2025057>.

POSTERS AND PRESENTATIONS

Paquin, D., Gross, L., Stewart, A., **Thai, G.** (2024) *Mathematically Modeling Chronic Myelogenous Leukemia*. Talk given in Jan 2025 at the Joint Mathematics Meetings in Seattle, WA.

Tully-Doyle, R., Adlin, L., **Thai, G.**, Tiscareno, S. (2023) *Graph Theoretic Interpretations of the Nevanlinna Representation*. Poster presented in Jan 2024 in the PME Poster Session, at the Joint Mathematics Meetings in San Francisco, CA.

Tully-Doyle, R., Adlin, L., **Thai, G.**, Tiscareno, S. (2023) *Graph Theoretic Interpretations of the Nevanlinna Representation*. Talk given in Nov 2023 at the 2nd CSU Mathematical Sciences Conference in Bakersfield, CA.

Lin, J., Ellwein, S., **Thai, G.** (2023) *Using Machine Learning to Improve Treatment Targeting in Farmer Training*. Poster presented in June 2023 at the Cal Poly BEACoN Research Symposium in San Luis Obispo, CA.

Liese, J., Klig, C., Lane, R., Moscot, M., **Thai, G.** (2022) *Characterizing Singular Graphs with Games*. Poster presented in June 2023 at the Cal Poly Bailey College Student Research Conference in San Luis Obispo, CA.

Liese, J., Klig, C., Lane, R., Moscot, M., **Thai, G.** (2022) *Characterizing Singular Graphs with Games*. Talk given in Nov 2022 at the 1st CSU Mathematical Sciences Conference in Northridge, CA.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Department of Mathematics

Purdue University

August 2025 - Present

- Quantitative Reasoning (MA 15555), two sections of 30 students. Fall 2026
- Multivariate Calculus (MA 26100), three sections of 40 students. Spring 2026
- Multivariate Calculus (MA 26100), three sections of 40 students. Fall 2025

Graduate Teaching Associate — Instructor of Record

Department of Mathematics

California Polytechnic State University

Jan 2024 - June 2025

- Precalculus Trigonometry (MATH 119), 35 students. Apr 2025 - June 2025
- Precalculus Trigonometry (MATH 119), 35 students. Apr 2024 - June 2024
- Precalculus Algebra (MATH 118), 35 students. Jan 2024 - Mar 2024

AWARDS AND SCHOLARSHIPS

- **Graduate Assistant Fellowship** Sep 2024
 - Awarded to fund master's thesis in collaboration with math faculty member.
- **Frost Undergraduate Research Fund** June 2024/2023/2022
 - Awarded to fund summer research projects in collaboration with mathematics faculty.
- **Edward van Duyne Scholarship** June 2024
 - Awarded to all class levels of Mathematics students with minimum 3.0 GPA.
- **Cal Poly BEACoN Research Grant** Jan 2023
 - Awarded to underrepresented students to fund research project with faculty member.